

**UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 205 – COMPUTATIONAL METHODS IN CHEMICAL ENGINEERING
Spring 2020**

Lecture:

MW: 10:00am – 11:15am

Room: EIB 124

Office Hours:

MW 2.15pm-3.15pm (Dr. Bilgin) (EIB 144) (by appointment)

Th 9am-11am (TA) (EIB 147)

Instructor:

Dr. Betul Bilgin, bbilgin@uic.edu

Teaching Assistant:

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Text:

Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB, by V.J. Law.
ISBN: 978-1-4665-7534-9, CRC Press, 2013.

Course Description:

Numerical and computational methods are introduced and applied to chemical engineering problems. Excel spreadsheets, VBA programming, algorithms and object-oriented concepts in chemical engineering.

Course Aims, Scope, Level:

Computer programming is playing essential role in science and engineering curriculum. UIC Chemical engineering curriculum aims at training the students in scientific and engineering programming using different programming languages. This course introduces fundamental formulation, methodology and techniques for numerical solution of chemical engineering problems via computing. The problems are introduced in the problem-specific context, including mass and energy balance, thermodynamics, fluid flow and heat transfer. This course mainly focuses on Excel and Visual Basic for Applications (VBA). The primary reason for using Excel is its availability. Other software program packages like MATLAB requires very expensive licenses for industry, therefore most likely, it will not be available to practicing engineers in industry.

Learning Objectives :

At the end of this course, a student should be able to:

- Formulate a chemical engineering problem as a mathematical model and select appropriate solution method.

- Write and modify VBA scripts for specific computational purposes and run them.
- Use Excel functionality to tabulate thermophysical property correlations and equations of state.
- Analyze 1d (linear) and 2d (bilinear) interpolation from data tables with VBA macros.
- Calculate vector coordinates for changes of basis vectors with Excel.
- Solve ordinary differential equations (ODE) with Excel.
- Solve optimization design problems with Excel Goal Seek and Solver functionalities and VBA macros.
- Solve partial differential equations (PDE).
- Carry out error analysis of data.
- Carry out regression analysis.

Course Outline:

- Ch. 1: Roots of a single nonlinear equation
- Ch. 2: Visual Basic for applications programming
- Ch. 3: Linear algebra and systems of linear equations
- Ch. 4: Numerical differentiation and integration
- Ch. 5: Ordinary differential equations (initial value problems)
- Ch. 6: Ordinary differential equations (boundary value problems)
- Ch. 8: Partial differential equations
- Ch. 7: Regression analysis and parameter estimation
- Ch. 9: Linear programming, nonlinear programming, nonlinear equations, and nonlinear regression using Solver

Supplemental Materials:

If you need to learn more about VBA programming, you may find many tutorials on the internet. You can also consult with the following websites:

[http://excelvbatutor.com/vba tutorial.html](http://excelvbatutor.com/vba%20tutorial.html)

[http://www.homeandlearn.org/basic mathematical operators.html](http://www.homeandlearn.org/basic%20mathematical%20operators.html)

<http://www.automateexcel.com/vba-code-examples>

Exams, Homework Assignments and Grading

There will be two midterm exams, and one final exam. All exams are required. There will be two 1 hour 15 minute-lectures per week (Mon-Wed). New material will be covered in the lecture. Students should stay current with the material and be prepared to answer questions. The teaching assistant is under no obligation to grade homework that is late, illegible, or sloppy. Scoring fewer than half of the total homework grade is grounds for failing the course. The homework is an important exercise necessary for learning the material and should be worked with care. Any evidence of copying will be addressed in accordance with the university's academic integrity policies. Assignments are due at midnight on the due date and must be submitted online through Blackboard. Late assignment (delivered before solutions are posted) will be penalized 50% on the grade of that assignment. Assignments turned in after solutions are posted will not be accepted. If there is only one late assignment throughout the semester, late penalty will be lifted from that assignment (i.e. half of the total credit for that assignment will be returned).

Grading Policy:

Course grades will be weighted according to the following grading formula.

Two exams (20% each)	40%
Final exam	25%
Homework problems	35%

The expectation is that all students will do well in this course if they actively participate in the course, do assignments on time and keep up with course materials.

Additional Information and Policies

1. EXAMS will be held during the class period. Makeup exams will not be provided except for health and weather-related emergencies.
2. Cheating on homework and exams will not be tolerated and appropriate action will be taken including (but not limited to) a failed grade or lowering of the final grade earned in the course.
3. Late Homework Policy: Homework can be turned in 1 day late without excuse for up to 50% credit.
4. Lecture Mobile Technology Policy will be given in 1st Lecture period.